AI-Native CRM — AI Guide

Use cases, configuration, and how to apply AI for every login and every workflow

Audience: administrators, AI governors, and every end user.
What this guide covers

1. **How the AI works** — the one governed gateway every AI feature runs through.
2. **Use cases & applications** — the full catalogue of AI actions, by module.
3. **How to configure AI** — tenant settings, providers, prompts, agents, and knowledge.
4. **How to apply AI for each login** — which AI each persona uses, and how to reach it.
5. **How to apply AI in workflows** — AI triggers and AI actions inside automations.
6. **Governance & safety** — review, redaction, logging, grounding, and escalation.

> **One principle underpins everything:** AI here is **governed, not magical**. Every call goes through a single gateway, draws its prompt from a managed registry (never hardcoded), checks your permissions, is logged, and — when the output is sensitive — **waits for a human to approve it** before it can be used. Answers that draw on your knowledge base **cite their sources** and **escalate instead of guessing** when they can't find a confident, approved answer.

Part 1 — How AI works here

Every AI capability is composed from five governed building blocks:

Block	What it is	Where you manage it
AI Gateway	The single entry point for all model calls — applies provider/model choice, rate limits, redaction, and logging.	`/admin` AI settings · `GET/PATCH /ai/settings`
Prompt Registry	Versioned, approvable prompt templates referenced by `templateKey`. Prompts are never hardcoded.	`/ai-prompts` · `/ai/prompts`
Agent Registry	Named agents with declared tools, data scope, and human-approval rules.	`/ai-agents` · `/ai/agents`
AI Actions	Predefined, per-module AI tasks (summaries, drafts, recommendations) with a `permitted` flag and review gate.	`/ai-actions` · `/ai/actions`
RAG / Knowledge	The approved knowledge that *grounds* answers, with retrieval you can inspect.	`/knowledge/*` · `/ai/knowledge/*`, `/ai/rag/retrieve`

The lifecycle of any AI request: `caller → permission check → Prompt Registry (resolve template) → AI Gateway (provider, rate limit, redaction) → execute → log (ai_usage_logs) → if sensitive: review queue → approved output usable`.

> **Live execution note.** The gateway and all governance are live; **live model output is deferred** until production provider credentials/execution backends are enabled — until then actions return governed placeholder output so you can build, permission, and review end to end safely.

Part 2 — AI use cases & applications (by module)

Every action below runs through the gateway, checks a module permission, and logs its run. Actions marked **(review)** are **sensitive** — their output is held as `pending_review` until a human approves it.

Module	AI actions
Leads	Lead summary · Lead score explanation · Follow-up email draft **(review)** · Lead qualification recommendation **(review)**
Accounts	Account brief · Relationship summary · Health indicator explanation
Opportunities	Opportunity summary · Deal risk analysis · Next-best-action **(review)** · Proposal draft outline **(review)**
Campaigns	Campaign plan generator **(review)** · Email copy generator **(review)** · Audience suggestion **(review)** · Campaign performance summary
Social	Caption generator **(review)** · Hashtag suggestion · Comment sentiment summary
Support	Ticket summary · Suggested response **(review)** · Knowledge article recommendation · Escalation summary
Customer Success	Onboarding plan generator **(review)** · Customer health summary · Churn risk explanation · Adoption recommendation **(review)** · QBR/EBR outline **(review)** · Renewal strategy suggestion **(review)**
Training	Lesson summary · Quiz generator · Learning path suggestion
Partners	Partner performance summary · Partner action plan **(review)** · Inactivity alert explanation
Resellers	Reseller performance summary · Reseller action plan **(review)** · Inactivity alert explanation

Plus three cross-cutting experiences:

- **AI Assistant / Ask AI** (`/ai-assistant` , `/ask-ai` , `/ai-help`) — a governed assistant for summaries, drafts, and grounded Q&A wherever you have `ai.use_ai` .
- **Customer-Query Bot** (`/portal` Ask AI; reviewed at `/customer-query`) — answers external customers **only** from approved knowledge, with citations, and escalates (often raising a ticket) when unsure.
- **RAG retrieval** (`/knowledge/rag-console`) — grounding for the above; preview exactly what the AI would retrieve for a query, with permission filtering.

Request types the gateway understands: `completion` , `summary` , `draft` , `classification` , `extraction` , `recommendation` (plus domain types: sales, presales, partners, support, customer_success, training, marketing, general).

Part 3 — How to configure AI

3.1 Tenant AI settings

The master controls for the whole tenant. **UI:** Administration → AI settings. **API:** `GET /ai/settings`, `PATCH /ai/settings` (needs `ai.configure` / `ai.manage_ai`).

Setting	What it does
`isEnabled`	Master on/off for all AI in the tenant.
`defaultProvider`	`openai` · `anthropic` · `azure_openai` · `local`.
`defaultModel`	e.g. `claude-opus-4-8`.
`rateLimitPerMinute`	Throttle to control cost/abuse.
`allowUserOverrides`	Whether callers may pick a different provider/model per request.
`redactionEnabled`	Strip sensitive data from prompts/outputs before they leave the gateway.
`loggingEnabled`	Record every request/response to `ai_usage_logs`.

```
PATCH /api/v1/ai/settings        # ai.configure
  { "isEnabled": true, "defaultProvider":
"anthropic", "defaultModel": "claude-opus-4-8",
    "rateLimitPerMinute": 60, "redactionEnabled":
true, "loggingEnabled": true,
    "allowUserOverrides": false }
GET /api/v1/ai/providers          # providers
available to this tenant
GET /api/v1/ai/templates          # prompt templates
the gateway can resolve
GET /api/v1/ai/logs   /ai/usage   # request logs and
usage (ai.view / view_dashboard / manage_ai)
```

Infrastructure-level (env, set by DevOps):
`AI_GATEWAY_ENABLED`, `AI_DEFAULT_PROVIDER`, `AI_DEFAULT_MODEL`, `AI_RATE_LIMIT_PER_MINUTE`, provider keys (`AI_OPENAI_API_KEY`, `AI_ANTHROPIC_API_KEY`, `AI_AZURE_OPENAI_API_KEY` / `ENDPOINT`, `AI_LOCAL_ENDPOINT`), `AI_EMBEDDING_MODEL`, `AI_VECTOR_BACKEND`, `AI_LOG_RETENTION_DAYS`. Adding real provider keys is what flips placeholder output into live model execution.

3.2 Prompt Registry — manage the prompts behind the AI

Prompts are versioned and approvable, so what the AI says is reviewable and controlled. **UI:** `/ai-prompts`. **API:** `/ai/prompts`.

```
POST /api/v1/ai/prompts
# create (ai.create / manage_ai)
  { "promptKey": "leads.followup_email", "name":
"Lead follow-up email",
    "module": "leads", "promptRole": "system",
"content": "...with {{variables}}...",
    "guardrails": { ... } }
POST /api/v1/ai/prompts/:id/versions
# new version (edit)
POST /api/v1/ai/prompts/:id/approval
# approve/reject (ai.approve)
POST
/api/v1/ai/prompts/:id/versions/:version/activate
# activate a version
```

Callers and workflows reference a prompt by its `templateKey` — change the template centrally and every caller updates.

3.3 Agent Registry — define what an agent may do

An agent bundles a purpose with **explicit guardrails**: which tools it may call, who may invoke it, what data it can see, and whether it needs human approval. **UI:** `/ai-agents`. **API:** `/ai/agents`.

Field	Meaning
`allowedTools`	The tools/functions the agent may use.
`allowedRoles`	Which roles may invoke it.
`dataAccessScope`	`own` · `team` · `module` · `tenant` — how widely it can read data.
`requiresHumanApproval`	Force a human gate on its outputs.
`status`	`draft` · `active` · `inactive`.
`loggingEnabled` / `escalationRules`	Logging and when to escalate to a human.

```
POST /api/v1/ai/agents          # ai.create /
manage_ai
  { "agentKey": "support.triage", "name": "Support
triage agent", "module": "support",
    "allowedTools": ["search_knowledge","summarize"],
"dataAccessScope": "module",
    "requiresHumanApproval": true, "status": "active"
}
```

3.4 Knowledge / RAG — ground the AI in *your* content

Grounded answers only use knowledge that is **approved + published**. **UI:** `/knowledge`, `/knowledge/upload`, `/knowledge/articles`, `/knowledge/rag-console`. **API:** `/ai/knowledge/*`, `/ai/rag/retrieve`.

1. Add a **source** and **upload documents** → process them (chunking) for retrieval.
2. Author **articles**; move them through **approved** → **published** (only then are they used by AI).
3. Use the **RAG console** (`POST /ai/rag/retrieve`) to preview what the AI retrieves and confirm permission filtering.
4. Watch **knowledge gaps** (`/ai/knowledge/gaps`) — recurring unanswered questions to write articles for.

Part 4 — How to apply AI for each login (persona)

Two ways any permitted user reaches AI: the **Ask AI** assistant (top-level or in-context), and **AI Actions** inside a module record (e.g. "Summarize", "Draft follow-up"). What each login sees depends on its `*.use_ai` / `ai.use_ai` permissions.

Login (persona)	AI they apply	H
SDR (`sales.development.representative`)	Lead summary, score explanation, follow-up draft *(review)*	C A r
Inside / Account Sales (`sales.executive` , `sales.manager` , …)	Opportunity summary, deal-risk, next-best-action *(review)*, account brief, proposal outline *(review)*	C o a s
Business Development (`business.development.*`)	Account/relationship summaries, proposal support	A a
Presales (`presales.*`)	Proposal draft outline *(review)*	A e r
Marketing (`marketing.*`)	Campaign plan *(review)*, email copy *(review)*, audience suggestion *(review)*, performance summary	C a r
Social (`social.media.marketing.*`)	Caption *(review)*, hashtags, comment sentiment	C
Support (`support.*`)	Ticket summary, suggested response *(review)*, article recommendation, escalation summary	C r q
Customer Success (`customer.success.*`)	Onboarding plan *(review)*, health/churn explanation, adoption *(review)*, QBR/EBR *(review)*, renewal strategy *(review)*	C –
Partner / Reseller Manager (`partner.manager` , `reseller.manager`)	Performance summary, action plan *(review)*, inactivity explanation	C r
Training / Knowledge (admin role)	Lesson summary, quiz generator, learning-path	

Login (persona)	AI they apply	H
	suggestion; author + publish grounding knowledge	
Executive Leadership (`executive.leadership`)	Executive insight summaries over dashboards	A
Customer Portal User (`customer.portal.user`)	Customer-query bot (grounded, cited, auto-escalating)	F
AI Administrator / Super Admin (`super.admin`)	Govern it all: settings, prompts, agents, review queue	p a

> Sign in on the demo (tenant `apar-elite`, password `AparDemo@2026!`) as any persona to see exactly which AI surfaces appear for that role.

Part 5 — How to apply AI in workflows

Workflows can be **driven by AI** and can **invoke AI** — automation plus the governed AI layer.

5.1 AI as a trigger

Two triggers fire on AI-derived signals:

- `ai_score_changed` — e.g. a lead's AI score crosses a threshold.
- `customer_health_changed` — e.g. a CS health score drops.

Use these to react automatically: *"when the AI lead score ≥ 80, notify the owner and create a task."*

5.2 AI as an action

Two workflow actions call the AI layer (they reference the registries from Part 3):

Action	What it does	References
`run_ai_prompt`	Runs a registered prompt template against the record's context.	a Prompt Registry `templateKey`
`run_ai_agent`	Invokes a registered agent (with its tools, scope, approval rule).	an Agent Registry `agentKey`

5.3 Build an AI-powered workflow (step by step)

UI: `/workflows` → New. **API** shown alongside.

1. **Trigger** — choose `ai_score_changed` (module `leads`). `POST /api/v1/workflows { "name":"Hot-lead AI assist","module":"leads","triggerType":"ai_score_changed" }`
2. **Condition** — only for hot leads: `score gte 80`. `"conditions":[{"field":"score","operator":"gte","value":80}]`
3. **AI action** — draft a follow-up via a registered prompt: `POST /api/v1/workflows/:id/actions { "actionType":"run_ai_prompt","sequence":1,"actionConfig":{"templateKey":"leads.followup_email"} }`
4. **Follow-on action** — notify the owner: `{ "actionType":"send_notification","sequence":2 }`.
5. **Test** — `POST /api/v1/workflows/:id/run { "context": { "score": 92 } }`, then check `GET /api/v1/workflows/:id/runs`.
6. **Activate** — `PATCH /api/v1/workflows/:id { "status":"active" }`.

> Because `leads.followup_email` is a *sensitive* draft, its output is held for **review** — the workflow produces a draft, not an autonomous send, keeping a human in the loop. Agents with `requiresHumanApproval` behave the same way.

Part 6 — Governance & safety (what protects you)

Control	Effect
Human-in-the-loop review	Sensitive action/agent output is `pending_review`; approve/reject at `/ai-actions` (`POST /ai/actions/runs/:runId/review`). Only approved output is usable.
Grounding + citations	Knowledge answers cite approved sources and won't fabricate.
Escalation	The customer bot escalates (raises a ticket) instead of guessing; gaps are logged for authors.
Redaction	`redactionEnabled` strips sensitive data at the gateway.
Rate limiting	`rateLimitPerMinute` caps usage.
Logging & usage	`loggingEnabled` records every run; review at `/ai/logs`, `/ai/usage` (retention `AI_LOG_RETENTION_DAYS`).
Permissioned data scope	Agents declare `own/team/module/tenant`; retrieval is permission-filtered.
Audit	AI configuration changes and approvals are written to the audit log.

Permissions quick reference

To…	You need
Use Ask AI / run an AI action	`ai.use_ai` or the module's `<module>.use_ai`
Review/approve a sensitive AI output	`ai.approve` / `ai.manage_ai`, or `<module>.approve`
Manage prompts & agents	`ai.manage_ai` (create/edit), `ai.approve` (approve prompt versions)
Change tenant AI settings	`ai.configure` / `ai.manage_ai`
See AI logs & usage	`ai.view`, `ai.view_dashboard`, `ai.manage_ai`, or `ai.configure`
Add AI steps to a workflow	`workflows.manage_workflow` / `workflows.configure` / `workflows.edit`

Quick start by goal

- **Turn AI on for the tenant:** Admin → AI settings → set provider/model, enable, set a rate limit, enable redaction + logging.
- **Let a team use AI:** grant that role `*.use_ai` (or `ai.use_ai`) in `/admin/rbac`.
- **Change what the AI says:** edit the template in `/ai-prompts`, approve, activate.
- **Automate with AI:** add a `run_ai_prompt` / `run_ai_agent` action to a workflow (Part 5).
- **Keep it safe:** leave sensitive actions on review; watch `/ai-actions` and `/ai/usage`.

Troubleshooting

Symptom	Cause	Fix
AI buttons missing	Role lacks `*.use_ai` / `ai.use_ai`, or `isEnabled` is off	Grant the permission; enable AI in settings
Output stuck "pending review"	It's a sensitive action	An approver acts in `/ai-actions`
Bot says it can't find an approved answer	No approved/published knowledge matched	It escalates; author an article for the logged gap
Placeholder-looking AI output	Live provider execution not yet enabled	Add provider keys / enable the execution backend
`403` calling an `/ai/*` endpoint	Token's role lacks the permission above	Use an admin/AI-governor token

---

*Companion documents: **Module Guide**, **Persona-Wise Workflow Guide**, and **Admin Configuration Guide**. Deeper design references live in `docs/ai/` (Architecture, Gateway, Governance, Prompt & Agent Registry, RAG, Customer-Query). Source of record: the live `/ai/*`, `/customer-query`, and `/workflows` APIs.*